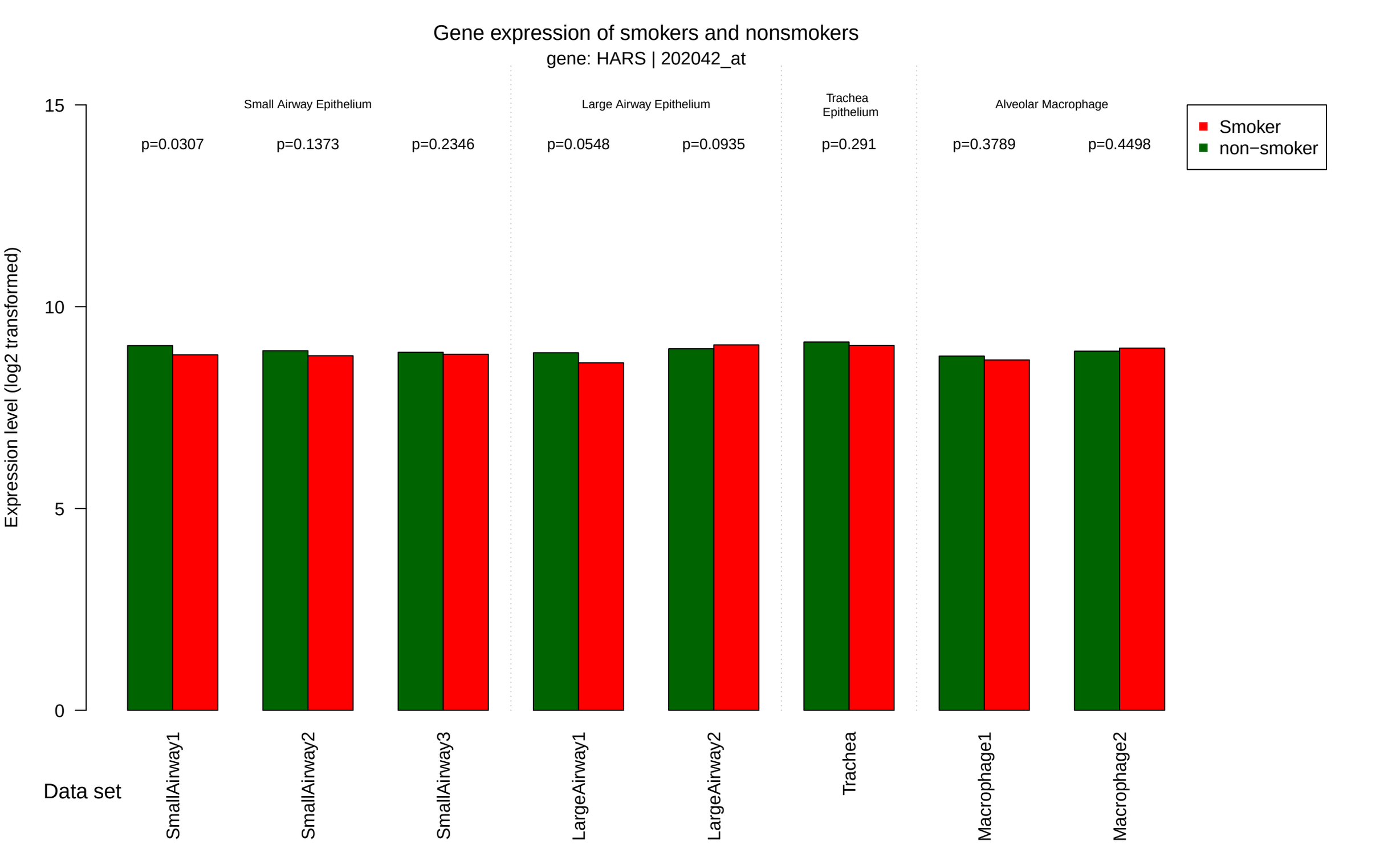
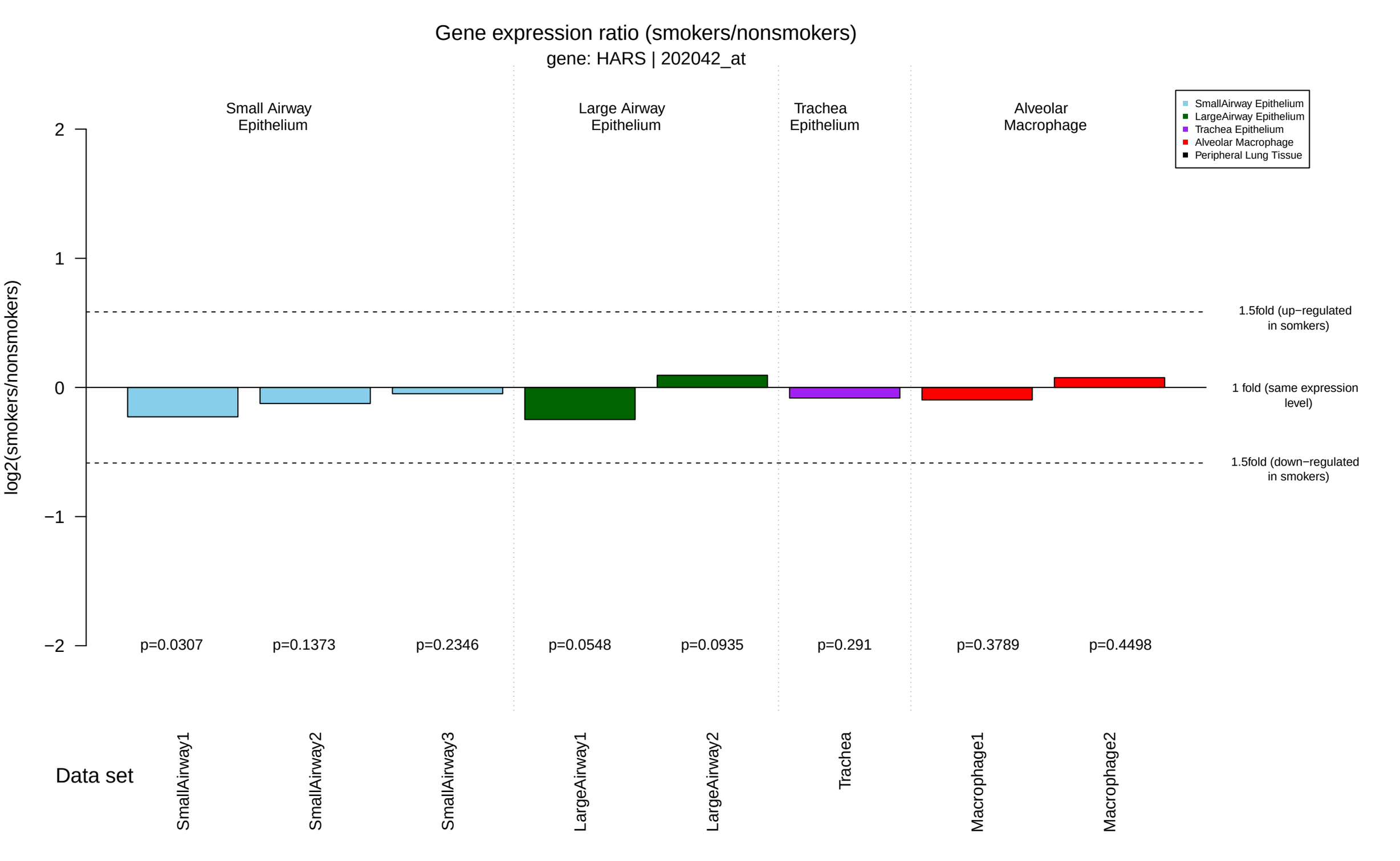
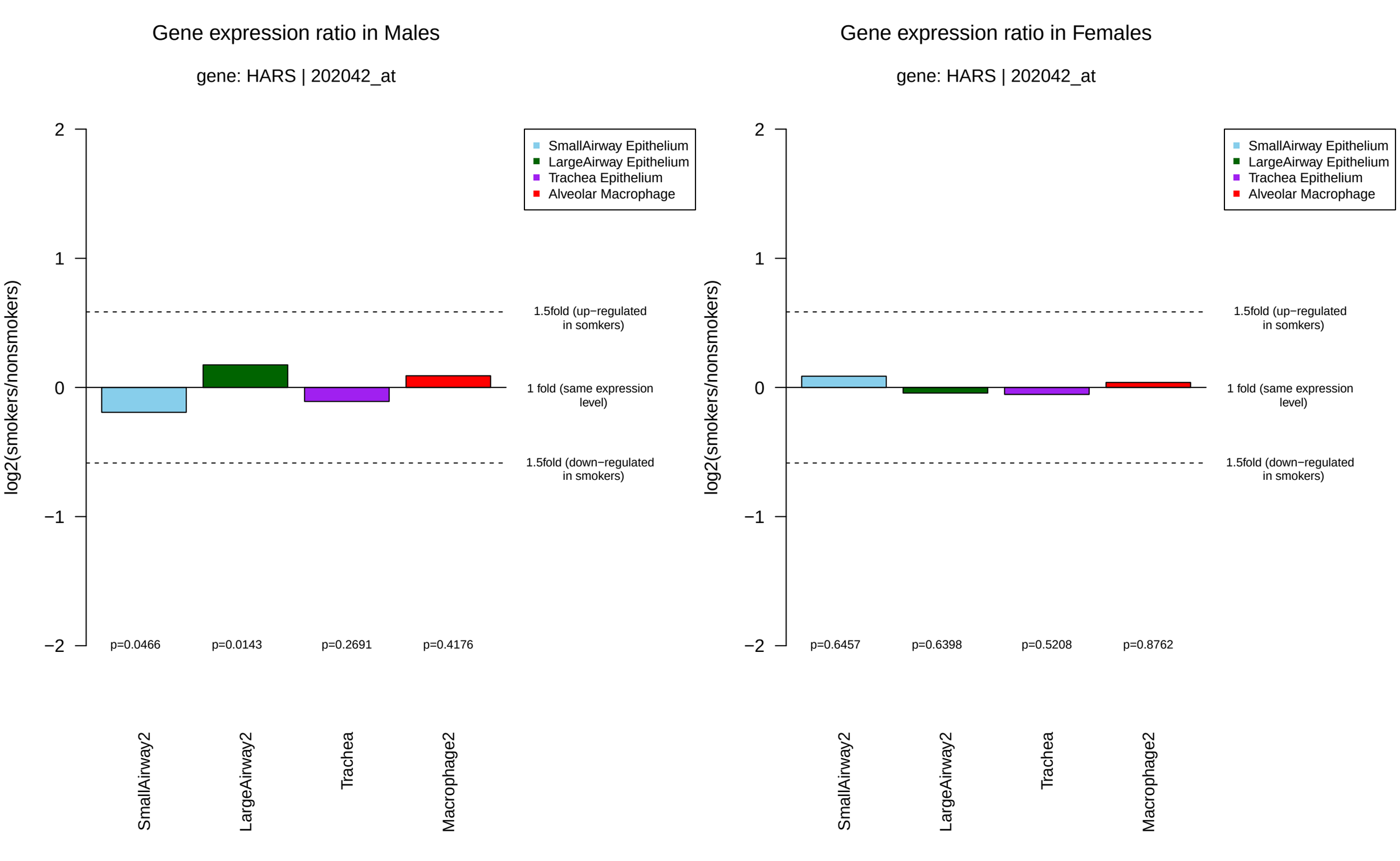
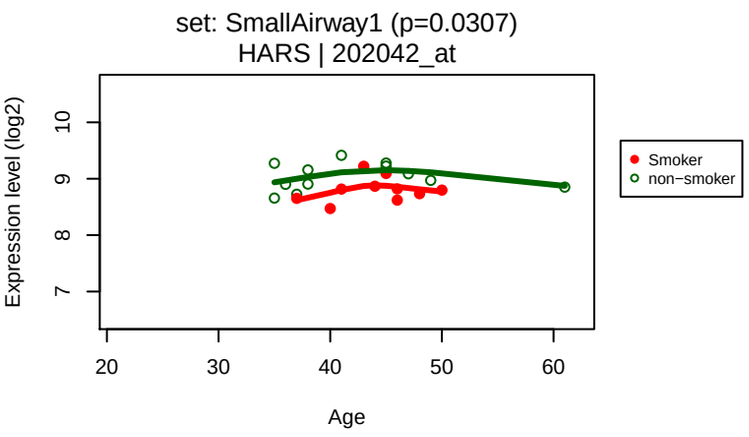
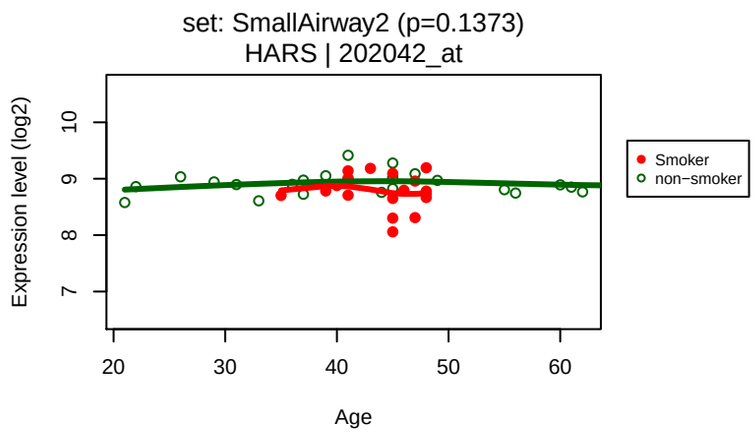
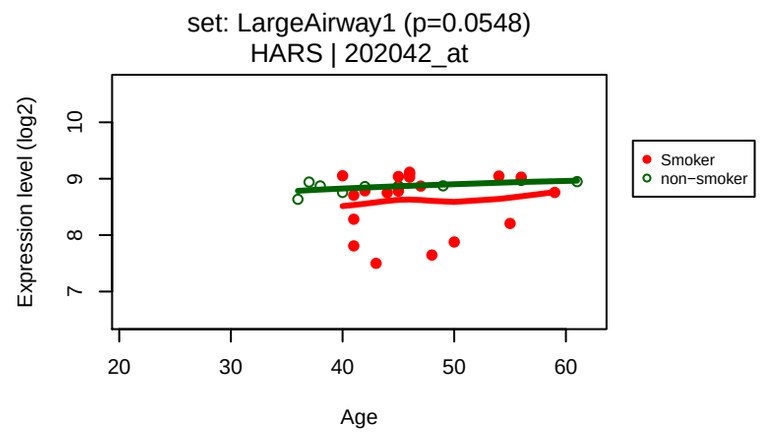

Gene expression of smokers and nonsmokers
gene: HARS | 202042_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0307
p=0.1373
p=0.2346
p=0.0548
p=0.0935
p=0.291
p=0.3789
p=0.4498
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

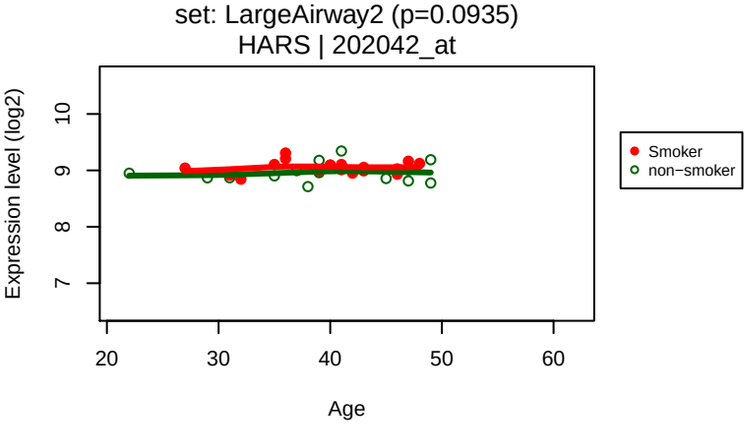

Gene expression ratio (smokers/nonsmokers)
gene: HARS | 202042_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0307
p=0.1373
p=0.2346
p=0.0548
p=0.0935
p=0.291
p=0.3789
p=0.4498
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

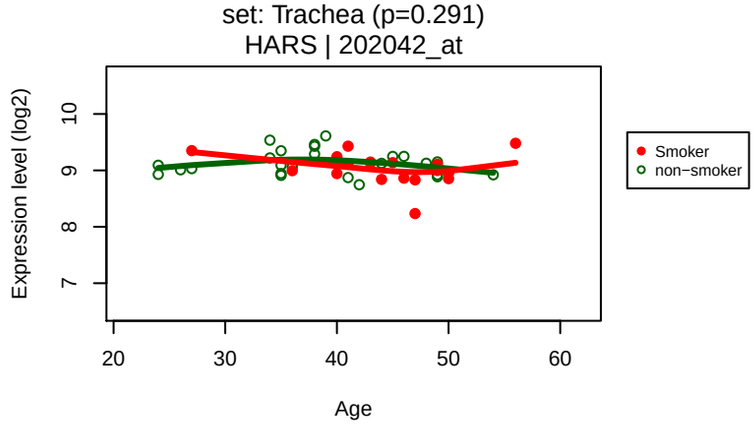

Gene expression ratio in Males
gene: HARS | 202042_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0466
p=0.0143
p=0.2691
p=0.4176
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HARS | 202042_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6457
p=0.6398
p=0.5208
p=0.8762
SmallAirway2
LargeAirway2
Trachea
Macrophage2

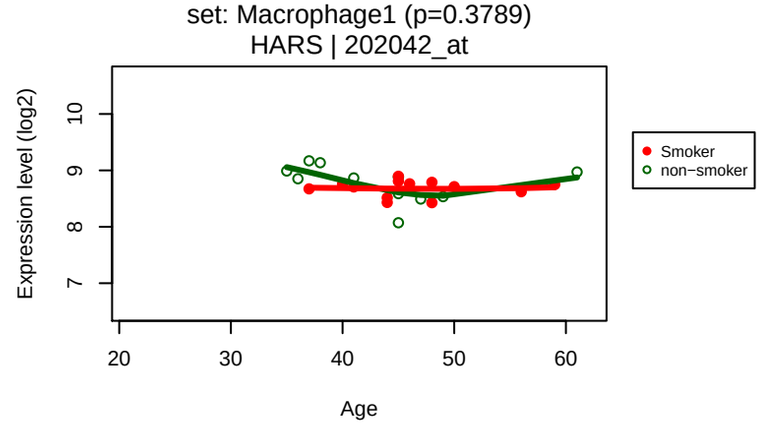

set: SmallAirway1 (p=0.0307)
HARS | 202042_at
Expression level (log2)
Age
Smoker
non−smoker

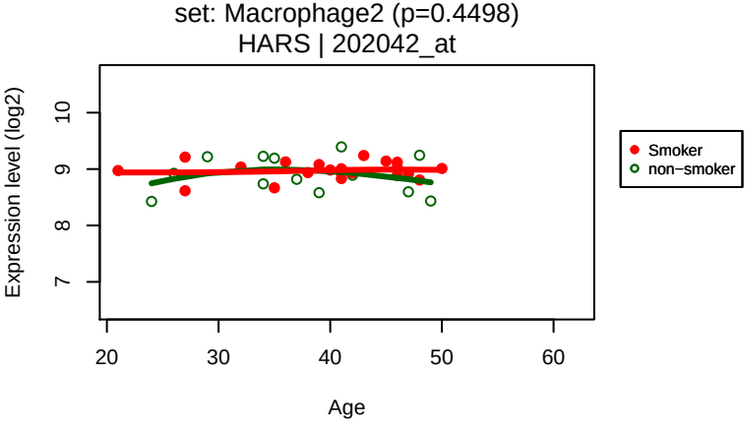

set: SmallAirway2 (p=0.1373)
HARS | 202042_at
Expression level (log2)
Age
Smoker
non−smoker

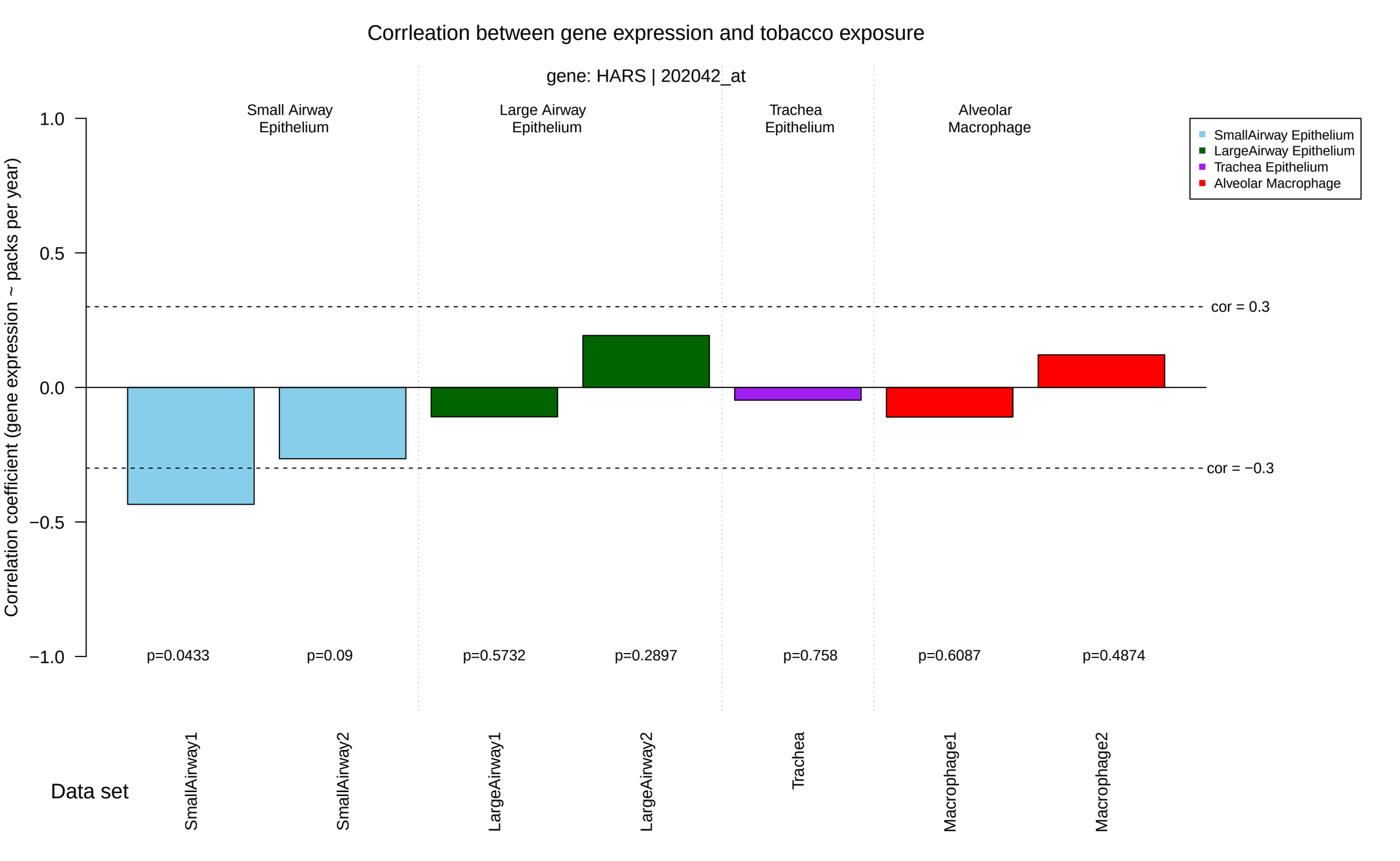

set: LargeAirway1 (p=0.0548)
HARS | 202042_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0935)
HARS | 202042_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.291)
HARS | 202042_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3789)
HARS | 202042_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4498)
HARS | 202042_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HARS | 202042_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0433
p=0.09
p=0.5732
p=0.2897
p=0.758
p=0.6087
p=0.4874
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2